Class 12 Mathematics

Chapter 10: Vector Algebra

Practice Set A: step-wise solutions

20 questions **47** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** PYQ-style — 1 mark

The value(s) of λ for which $\lambda(2\hat{i}-\hat{j}+2\hat{k})$ is a unit vector is/are:

(A) $\pm\frac{1}{3}$

(B) ± 3

(C) $\frac{1}{3}$ only

(D) $\pm\frac{1}{9}$

Formula / idea: $\vec{v}$ is a unit vector when $|\vec{v}|=1$; and $|\lambda\vec{a}|=|\lambda|\,|\vec{a}|$.

1 A unit vector has magnitude 1.
$|\lambda(2\hat{i}-\hat{j}+2\hat{k})|=1$

2 Take $|\lambda|$ outside the magnitude.
$|\lambda|\sqrt{4+1+4}=1 \;\Rightarrow\; 3|\lambda|=1$

3 Solve for λ. ✓ 1
$|\lambda|=\frac{1}{3} \;\Rightarrow\; \lambda=\pm\frac{1}{3}$

4 Hence the answer is option (A).

Answer **(A)** $\pm\frac{1}{3}$

Watch out: Choosing (C): $\lambda=-\frac{1}{3}$ also gives magnitude 1, so both signs work.

2 MCQ CBSE SQP 2024-25 1 mark

If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $|\vec{a} + \vec{b}| = 5$, then $|\vec{a} - \vec{b}| =$

(A) 3 **(B)** 4

(C) 5 **(D)** 8

Formula / idea: $|\vec{a} \pm \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 \pm 2\,\vec{a} \cdot \vec{b}$.

1 Expand $|\vec{a} + \vec{b}|^2$ and put the values.
$25 = 9 + 16 + 2\,\vec{a} \cdot \vec{b}$

2 Solve for the dot product.
$\vec{a} \cdot \vec{b} = 0$

3 Expand $|\vec{a} - \vec{b}|^2$. ✓ 1
$|\vec{a} - \vec{b}|^2 = 9 + 16 - 2(0) = 25 \;\Rightarrow\; |\vec{a} - \vec{b}| = 5$

4 Hence the answer is option (C).

Answer **(C)** 5

Watch out: Subtracting magnitudes: $|\vec{a} - \vec{b}| \neq |\vec{a}| - |\vec{b}|$; square and expand instead.

3 MCQ CBSE SQP 2024-25 1 mark

The value of α for which the angle between $\vec{p} = 2\alpha^2\hat{i} - 3\alpha\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + \hat{j} + \alpha\hat{k}$ is obtuse, is:

(A) $\mathbb{R} - [0, 1]$ **(B)** $(0, 1)$

(C) $[0, \infty)$ **(D)** $[1, \infty)$

Formula / idea: The angle between $\vec{p}$ and $\vec{q}$ is obtuse when $\vec{p} \cdot \vec{q} < 0$.

1 Find the dot product.
$\vec{p} \cdot \vec{q} = 2\alpha^2 - 3\alpha + \alpha = 2\alpha(\alpha - 1)$

2 For an obtuse angle the dot product is negative.
$2\alpha(\alpha - 1) < 0$

3 A product is negative when the factors have opposite signs. ✓ 1
$0 < \alpha < 1 \;\Rightarrow\; \alpha \in (0, 1)$

4 Hence the answer is option (B).
$\vec{p}$ and $\vec{q}$ are never opposite, so $\theta \neq \pi$.

Answer **(B)** $(0, 1)$

Watch out: Choosing (A): $\mathbb{R} - [0, 1]$ is where $\vec{p} \cdot \vec{q} > 0$, the acute case.

4 MCQ PYQ-style 1 mark

If $|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = 400$ and $|\vec{a}| = 5$, then $|\vec{b}|$ is equal to:

(A) 4 **(B)** 16

(C) 80 **(D)** 400

Formula / idea: $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$.

1 Use $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$ and $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$.
$|\vec{a}|^2|\vec{b}|^2(\sin^2\theta + \cos^2\theta) = 400$

2 Since $\sin^2\theta + \cos^2\theta = 1$, put $|\vec{a}| = 5$. ✓ 1
$25\,|\vec{b}|^2 = 400 \Rightarrow |\vec{b}|^2 = 16 \Rightarrow |\vec{b}| = 4$

3 Hence the answer is option (A).

Answer **(A)** 4

Watch out: Stopping at $|\vec{b}|^2 = 16$ and choosing (B).

5 MCQ CBSE SQP 2025-26 1 mark

A bird flies through a distance in a straight line given by the vector $\hat{i} + 2\hat{j} + \hat{k}$. A man standing beside a straight metro rail track given by $\vec{r} = (3 + \lambda)\hat{i} + (2\lambda - 1)\hat{j} + 3\lambda\hat{k}$ is observing the bird. The projected length of its flight on the metro track is:

(A) $\dfrac{6}{\sqrt{14}}$ units

(B) $\dfrac{14}{\sqrt{6}}$ units

(C) $\dfrac{8}{\sqrt{14}}$ **units**

(D) $\dfrac{5}{\sqrt{6}}$ units

Formula / idea: Projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$.

1 The track runs along the coefficient of λ.
$\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}, \quad |\vec{b}| = \sqrt{14}$

2 Dot the flight vector with $\vec{b}$.
$(\hat{i} + 2\hat{j} + \hat{k}) \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 1 + 4 + 3 = 8$

3 Divide by $|\vec{b}|$. ✓ 1
$\text{Projection} = \dfrac{8}{\sqrt{14}}$ units

4	Hence the answer is option (C).	

Answer **(C)** $\frac{8}{\sqrt{14}}$ **units**

Watch out: Taking $3\hat{i} - \hat{j}$ (the $\lambda = 0$ point) as the track direction; the direction is the λ-part.

6 **Assertion-Reason** CBSE SQP 2025-26 1 mark

Assertion (A): Given two non-zero vectors $\vec{a}$ and $\vec{b}$. If $\vec{r}$ is another non-zero vector such that $\vec{r} \times (\vec{a} + \vec{b}) = \vec{0}$, then $\vec{r}$ is perpendicular to $\vec{a} \times \vec{b}$. **Reason (R):** The vector $(\vec{a} + \vec{b})$ is perpendicular to the plane of $\vec{a}$ and $\vec{b}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $\vec{r} \times \vec{v} = \vec{0}$ means $\vec{r}$ is collinear with $\vec{v}$; $\vec{a} \times \vec{b}$ is perpendicular to the plane of $\vec{a}, \vec{b}$.

1	Cross product zero means the vectors are collinear. $\vec{r} \times (\vec{a} + \vec{b}) = \vec{0} \Rightarrow \vec{r} = \lambda(\vec{a} + \vec{b})$	
2	So $\vec{r}$ lies in the plane of $\vec{a}$ and $\vec{b}$; $\vec{a} \times \vec{b}$ is normal to it. $\vec{r} \cdot (\vec{a} \times \vec{b}) = \lambda(\vec{a} + \vec{b}) \cdot (\vec{a} \times \vec{b}) = 0$	
3	Hence $\vec{r} \perp (\vec{a} \times \vec{b})$. **A is true.**	
4	$\vec{a} + \vec{b}$ is a combination of $\vec{a}, \vec{b}$, so it lies in their plane. **R is false.**	
5	Hence the correct option is (C).	✓ 1

Answer **(C) A is true, but R is false**

Watch out: Mixing up $\vec{a} + \vec{b}$ (in the plane) with $\vec{a} \times \vec{b}$ (perpendicular to the plane).

7 **Short answer** CBSE SQP 2025-26 2 marks

The two vectors $\hat{i}+\hat{j}+\hat{k}$ and $3\hat{i}-\hat{j}+3\hat{k}$ represent the two sides OA and OB, respectively of a $\triangle OAB$, where O is the origin. The point P lies on AB such that OP is a median. Find the area of the parallelogram formed by the two adjacent sides as OA and OP.

Formula / idea: Area of parallelogram with adjacent sides $\vec{a}, \vec{b} = |\vec{a} \times \vec{b}|$.

1 P is the mid-point of AB.
$\overrightarrow{OP} = \dfrac{\overrightarrow{OA}+\overrightarrow{OB}}{2} = \dfrac{4\hat{i}+0\hat{j}+4\hat{k}}{2} = 2\hat{i}+2\hat{k}$ ✓ ½

2 Write the formula for the area.
$\text{Area} = |\overrightarrow{OA} \times \overrightarrow{OP}|$

3 Find the cross product.
$\overrightarrow{OA} \times \overrightarrow{OP} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & 0 & 2 \end{vmatrix} = 2\hat{i}+0\hat{j}-2\hat{k}$ ✓ ½

4 Find its magnitude.
$|2\hat{i}-2\hat{k}| = \sqrt{4+0+4} = 2\sqrt{2}$ ✓ ½

5 State the area with units.
$\text{Area} = 2\sqrt{2}$ sq units ✓ ½

Answer **Area $= 2\sqrt{2}$ sq units**

Watch out: Using $\overrightarrow{OB}$ instead of $\overrightarrow{OP}$, or halving the answer as if it were a triangle.

Level 2

16 marks

8 **Short answer** PYQ-style 2 marks

Find a vector of magnitude 5 units in the direction of the resultant of the vectors $\vec{a} = 2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{b} = \hat{i}-2\hat{j}+\hat{k}$.

Formula / idea: Vector of magnitude m along $\vec{v} = m\,\hat{v} = m\dfrac{\vec{v}}{|\vec{v}|}$.

1 Find the resultant.
$\vec{a}+\vec{b} = 3\hat{i}+\hat{j}+0\hat{k}$ ✓ ½

2 Find its magnitude.
$|\vec{a}+\vec{b}| = \sqrt{9+1+0} = \sqrt{10}$ ✓ ½

3 Multiply the unit vector by 5. ✓ ½

$$5 \cdot \frac{3\hat{i}+\hat{j}}{\sqrt{10}} = \frac{\sqrt{10}}{2}(3\hat{i}+\hat{j})$$

$$\frac{5}{\sqrt{10}} = \frac{\sqrt{10}}{2}$$

4 Write the required vector. ✓ ½

$$\frac{3\sqrt{10}}{2}\hat{i} + \frac{\sqrt{10}}{2}\hat{j}$$

Answer $\frac{3\sqrt{10}}{2}\hat{i} + \frac{\sqrt{10}}{2}\hat{j}$

Watch out: Multiplying $\vec{a}+\vec{b}$ by 5 directly; first make it a unit vector.

9 **Short answer** PYQ-style 2 marks

Find a unit vector perpendicular to the plane ABC, where A, B and C are the points $(3,-1,2)$, $(1,-1,-3)$ and $(4,-3,1)$ respectively.

Formula / idea: $\overrightarrow{AB} \times \overrightarrow{AC}$ is perpendicular to the plane ABC.

1 Find two sides of the triangle. ✓ ½

$$\overrightarrow{AB} = -2\hat{i} + 0\hat{j} - 5\hat{k}, \quad \overrightarrow{AC} = \hat{i} - 2\hat{j} - \hat{k}$$

2 Take their cross product. ✓ ½

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 0 & -5 \\ 1 & -2 & -1 \end{vmatrix} = -10\hat{i} - 7\hat{j} + 4\hat{k}$$

3 Find its magnitude. ✓ ½

$$\sqrt{100+49+16} = \sqrt{165}$$

4 Divide by the magnitude; both directions work. ✓ ½

$$\hat{n} = \pm\frac{1}{\sqrt{165}}(-10\hat{i} - 7\hat{j} + 4\hat{k})$$

Answer $\pm\frac{1}{\sqrt{165}}(-10\hat{i} - 7\hat{j} + 4\hat{k})$

Watch out: Expanding the $\hat{j}$ term with a $+$ sign; the middle term of the determinant is subtracted.

10 **Short answer** CBSE SQP 2024-25 2 marks

A person standing at $O(0,0,0)$ is watching an aeroplane which is at the coordinate point $A(4,0,3)$. At the same time he saw a bird at the coordinate point $B(0,0,1)$. Find the angles which $\overrightarrow{BA}$ makes with the x, y and z axes.

Formula / idea: If $\vec{r} = a\hat{i} + b\hat{j} + c\hat{k}$, then $\cos\alpha = \frac{a}{|\vec{r}|}$, $\cos\beta = \frac{b}{|\vec{r}|}$, $\cos\gamma = \frac{c}{|\vec{r}|}$.

1 Find $\overrightarrow{BA}$ as head minus tail. ✓ ½

$\overrightarrow{BA} = (4-0)\hat{i} + (0-0)\hat{j} + (3-1)\hat{k} = 4\hat{i} + 0\hat{j} + 2\hat{k}$

2 Find its magnitude. ✓ ½

$|\overrightarrow{BA}| = \sqrt{16+0+4} = 2\sqrt{5}$

3 Find the direction cosines. ✓ ½

$\cos\alpha = \frac{4}{2\sqrt{5}} = \frac{2}{\sqrt{5}}, \quad \cos\beta = 0, \quad \cos\gamma = \frac{2}{2\sqrt{5}} = \frac{1}{\sqrt{5}}$

4 Write the angles. ✓ ½

$\alpha = \cos^{-1}\frac{2}{\sqrt{5}}, \quad \beta = \frac{\pi}{2}, \quad \gamma = \cos^{-1}\frac{1}{\sqrt{5}}$

Answer $\cos^{-1}\frac{2}{\sqrt{5}}, \frac{\pi}{2}, \cos^{-1}\frac{1}{\sqrt{5}}$

Watch out: Using $\overrightarrow{OA}$ instead of $\overrightarrow{BA}$; the question is about the line from the bird to the plane.

11 **Short answer** PYQ-style 3 marks

Express the vector $\vec{a} = 5\hat{i} - 2\hat{j} + 5\hat{k}$ as the sum of two vectors such that one is parallel to the vector $\vec{b} = 3\hat{i} + \hat{k}$ and the other is perpendicular to $\vec{b}$.

Formula / idea: Parallel to $\vec{b}$ means $\lambda\vec{b}$; perpendicular to $\vec{b}$ means dot product with $\vec{b}$ is 0.

1 Let the part parallel to $\vec{b}$ be $\lambda\vec{b}$. ✓ ½

$\vec{a}_1 = \lambda\vec{b} = 3\lambda\hat{i} + \lambda\hat{k}$

2 The other part is what is left over. ✓ ½

$\vec{a}_2 = \vec{a} - \lambda\vec{b} = (5-3\lambda)\hat{i} - 2\hat{j} + (5-\lambda)\hat{k}$

3 $\vec{a}_2 \perp \vec{b}$, so their dot product is 0. ✓ ½

$\vec{a}_2 \cdot \vec{b} = 3(5-3\lambda) + 0(-2) + 1(5-\lambda) = 0$

4 Simplify and solve. ✓ ½

$20 - 10\lambda = 0 \Rightarrow \lambda = 2$

5 Put $\lambda = 2$ to get the two parts. ✓ ½

$\vec{a}_1 = 6\hat{i} + 2\hat{k}, \quad \vec{a}_2 = -\hat{i} - 2\hat{j} + 3\hat{k}$

6 Hence the required sum is
$\vec{a} = (6\hat{i} + 2\hat{k}) + (-\hat{i} - 2\hat{j} + 3\hat{k})$
Check: $(-\hat{i} - 2\hat{j} + 3\hat{k}) \cdot (3\hat{i} + \hat{k}) = -3 + 3 = 0$. ✓ ½

Answer $\vec{a} = (6\hat{i} + 2\hat{k}) + (-\hat{i} - 2\hat{j} + 3\hat{k})$

Watch out: Reading $\vec{b}$ as having a $\hat{j}$ term; its $\hat{j}$-component is 0.

12 **Short answer** CBSE SQP 2024-25 3 marks

An ant is moving along the vector $\vec{l_1} = \hat{i} - 2\hat{j} + 3\hat{k}$. Few sugar crystals are kept along the vector $\vec{l_2} = 3\hat{i} - 2\hat{j} + \hat{k}$ which is inclined at an angle θ with the vector $\vec{l_1}$. Then find the angle θ. Also find the scalar projection of $\vec{l_1}$ on $\vec{l_2}$.

Formula / idea: $\cos\theta = \dfrac{\vec{l_1} \cdot \vec{l_2}}{|\vec{l_1}||\vec{l_2}|}$; projection of $\vec{l_1}$ on $\vec{l_2} = \dfrac{\vec{l_1} \cdot \vec{l_2}}{|\vec{l_2}|}$.

1 Find the dot product. ✓ ½
$\vec{l_1} \cdot \vec{l_2} = 1(3) + (-2)(-2) + 3(1) = 10$

2 Find the magnitudes. ✓ ½
$|\vec{l_1}| = \sqrt{1+4+9} = \sqrt{14}, \quad |\vec{l_2}| = \sqrt{9+4+1} = \sqrt{14}$

3 Use the angle formula. ✓ ½
$\cos\theta = \dfrac{10}{\sqrt{14}\sqrt{14}} = \dfrac{10}{14} = \dfrac{5}{7}$

4 Write the angle. ✓ ½
$\theta = \cos^{-1}\dfrac{5}{7}$

5 Find the scalar projection of $\vec{l_1}$ on $\vec{l_2}$. ✓ 1
$\dfrac{\vec{l_1} \cdot \vec{l_2}}{|\vec{l_2}|} = \dfrac{10}{\sqrt{14}} = \dfrac{5\sqrt{14}}{7}$

Answer $\theta = \cos^{-1}\dfrac{5}{7}$; **projection** $= \dfrac{10}{\sqrt{14}} = \dfrac{5\sqrt{14}}{7}$

Watch out: Dividing by $|\vec{l_1}|$ for the projection on $\vec{l_2}$; divide by the length of the vector you project onto.

13 **Short answer** PYQ-style 2 marks

If $|\vec{a}| = 3$ and $|\vec{b}| = 4$, find the value(s) of λ for which the vectors $\vec{a} + \lambda\vec{b}$ and $\vec{a} - \lambda\vec{b}$ are perpendicular.

Formula / idea: Two vectors are perpendicular if and only if their dot product is 0.

1 Perpendicular vectors have dot product zero. ✓ ½
$(\vec{a} + \lambda\vec{b}) \cdot (\vec{a} - \lambda\vec{b}) = 0$

2 Expand; the middle terms cancel since $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$. ✓ ½
$|\vec{a}|^2 - \lambda^2|\vec{b}|^2 = 0$

3 Put the magnitudes. ✓ ½
$9 - 16\lambda^2 = 0 \Rightarrow \lambda^2 = \dfrac{9}{16}$

4 Take the square root. ✓ ½
$\lambda = \pm\dfrac{3}{4}$

Answer $\lambda = \pm\dfrac{3}{4}$

Watch out: Giving only $\lambda = \frac{3}{4}$; the negative root also works.

14 **Short answer** PYQ-style 2 marks

During a cricket match, the positions of the bowler, the wicket-keeper and the leg-slip fielder are in a line, given by $\vec{B} = 2\hat{i} + 8\hat{j}$, $\vec{W} = 6\hat{i} + 12\hat{j}$ and $\vec{F} = 12\hat{i} + 18\hat{j}$ respectively. Find the ratio in which the wicket-keeper divides the line segment joining the bowler and the fielder.

Formula / idea: Section formula: the point dividing $\vec{a}$, $\vec{b}$ internally in $m : n$ is $\dfrac{m\vec{b} + n\vec{a}}{m + n}$.

1 Let W divide BF in the ratio $k : 1$. ✓ ½
$\vec{W} = \dfrac{k\vec{F} + \vec{B}}{k + 1}$

2 Substitute the position vectors. ✓ ½
$6\hat{i} + 12\hat{j} = \dfrac{(12k + 2)\hat{i} + (18k + 8)\hat{j}}{k + 1}$

3 Compare the $\hat{i}$ components. ✓ ½
$12k + 2 = 6k + 6 \Rightarrow k = \dfrac{2}{3}$

4 Check with $\hat{j}$ and state the ratio. ✓ ½

$$\frac{18 \cdot \frac{2}{3} + 8}{\frac{5}{3}} = 12 \checkmark \Rightarrow \text{ratio } 2:3 \text{ internally}$$

Answer $2:3$ **internally**

Watch out: Writing $\frac{k\vec{B} + \vec{F}}{k+1}$; the ratio's first number goes with the far end $\vec{F}$.

Level 3

23 marks

15 **Short answer** PYQ-style 3 marks

If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular vectors of equal magnitudes, show that the vector $\vec{a} + \vec{b} + \vec{c}$ is equally inclined to $\vec{a}$, $\vec{b}$ and $\vec{c}$.

Formula / idea: Mutually perpendicular: $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$; $\cos\theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}||\vec{v}|}$.

1 Write the given conditions. ✓ ½

$$|\vec{a}| = |\vec{b}| = |\vec{c}| = p, \quad \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$$

2 Find $|\vec{a} + \vec{b} + \vec{c}|^2$; the dot-product terms are zero. ✓ ½

$$|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 = 3p^2$$

3 So the magnitude is ✓ ½

$$|\vec{a} + \vec{b} + \vec{c}| = \sqrt{3}\,p$$

4 Let α be the angle with $\vec{a}$. ✓ ½

$$\cos\alpha = \frac{(\vec{a} + \vec{b} + \vec{c}) \cdot \vec{a}}{\sqrt{3}\,p \cdot p} = \frac{p^2 + 0 + 0}{\sqrt{3}\,p^2} = \frac{1}{\sqrt{3}}$$

5 In the same way, for the angles β with $\vec{b}$ and γ with $\vec{c}$: ✓ ½

$$\cos\beta = \frac{1}{\sqrt{3}}, \quad \cos\gamma = \frac{1}{\sqrt{3}}$$

6 So $\alpha = \beta = \gamma$. Hence $\vec{a} + \vec{b} + \vec{c}$ is equally inclined to $\vec{a}, \vec{b}, \vec{c}$. ✓ ½

$$\alpha = \beta = \gamma = \cos^{-1}\frac{1}{\sqrt{3}}$$

Answer **Each angle is** $\cos^{-1}\frac{1}{\sqrt{3}}$**, so** $\vec{a} + \vec{b} + \vec{c}$ **is equally inclined to** $\vec{a}, \vec{b}, \vec{c}$**.**

Watch out: Writing $|\vec{a} + \vec{b} + \vec{c}| = 3p$; magnitudes do not add, so square first.

16 **Long answer** PYQ-style 5 marks

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$. Find a vector $\vec{c}$ such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$.

Formula / idea: Write $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$ and compare components of equal vectors.

1 Let the required vector be
$\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$ ✓ ½

2 Find $\vec{a} \times \vec{c}$.
$$\vec{a} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ x & y & z \end{vmatrix} = (z-y)\hat{i} + (x-z)\hat{j} + (y-x)\hat{k}$$ ✓ 1

3 Put $\vec{a} \times \vec{c} = \vec{b} = 0\hat{i} + \hat{j} - \hat{k}$ and compare components.
$z - y = 0, \quad x - z = 1, \quad y - x = -1$ ✓ 1

4 Use $\vec{a} \cdot \vec{c} = 3$.
$x + y + z = 3$ ✓ ½

5 From the first two equations, write x and y in terms of z.
$y = z, \quad x = z + 1$
The third equation follows from these two. ✓ ½

6 Substitute in $x + y + z = 3$.
$(z+1) + z + z = 3 \Rightarrow z = \frac{2}{3}$ ✓ ½

7 Find x and y.
$y = \frac{2}{3}, \quad x = \frac{5}{3}$ ✓ ½

8 Write the required vector.
$\vec{c} = \frac{5}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}$ ✓ ½

Answer $\vec{c} = \frac{5}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k} = \frac{1}{3}(5\hat{i} + 2\hat{j} + 2\hat{k})$

Watch out: Getting the $\hat{j}$ component of $\vec{a} \times \vec{c}$ as $z - x$; the minus sign before $\hat{j}$ makes it $x - z$.

17 **Short answer** PYQ-style 3 marks

The scalar product of the vector $\hat{i} + \hat{j} + \hat{k}$ with a unit vector along the sum of the vectors $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\lambda\hat{i} + 2\hat{j} + 3\hat{k}$ is equal to one. Find the value of λ.

Formula / idea: Unit vector along $\vec{s} = \frac{\vec{s}}{|\vec{s}|}$.

1 Find the sum of the two vectors.
$\vec{s} = (2 + \lambda)\hat{i} + 6\hat{j} - 2\hat{k}$ ✓ ½

2 Find its magnitude. ✓ ½

$|\vec{s}| = \sqrt{(2+\lambda)^2 + 36 + 4} = \sqrt{\lambda^2 + 4\lambda + 44}$

3 Write the unit vector along $\vec{s}$. ✓ ½

$\hat{s} = \dfrac{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{\lambda^2 + 4\lambda + 44}}$

4 Its dot product with $\hat{i} + \hat{j} + \hat{k}$ is 1. ✓ ½

$\dfrac{(2+\lambda) + 6 - 2}{\sqrt{\lambda^2 + 4\lambda + 44}} = 1 \;\Rightarrow\; \lambda + 6 = \sqrt{\lambda^2 + 4\lambda + 44}$

5 Square both sides. ✓ ½

$\lambda^2 + 12\lambda + 36 = \lambda^2 + 4\lambda + 44$

6 Solve for λ. ✓ ½

$8\lambda = 8 \;\Rightarrow\; \lambda = 1$

Check: $\lambda + 6 = 7 > 0$, so squaring added no extra root.

Answer $\lambda = 1$

Watch out: Dotting with the sum itself; first divide the sum by its magnitude to get the unit vector.

18 **Short answer** PYQ-style 3 marks

Using vectors, find the value of k such that the points $A(k, -10, 3)$, $B(1, -1, 3)$ and $C(3, 5, 3)$ are collinear.

Formula / idea: A, B, C are collinear $\iff$ area of $\triangle ABC = 0 \iff \overrightarrow{AB} \times \overrightarrow{AC} = \vec{0}$.

1 Find two sides from A. ✓ ½

$\overrightarrow{AB} = (1-k)\hat{i} + 9\hat{j} + 0\hat{k}, \quad \overrightarrow{AC} = (3-k)\hat{i} + 15\hat{j} + 0\hat{k}$

2 Collinear points form a triangle of zero area. ✓ ½

$\overrightarrow{AB} \times \overrightarrow{AC} = \vec{0}$

3 Find the cross product. ✓ ½

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1-k & 9 & 0 \\ 3-k & 15 & 0 \end{vmatrix} = \big[15(1-k) - 9(3-k)\big]\hat{k}$$

4 Simplify the $\hat{k}$ component. ✓ ½

$15 - 15k - 27 + 9k = -12 - 6k$

5 Set it equal to zero. ✓ ½

$-12 - 6k = 0 \;\Rightarrow\; k = -2$

6 Hence the points are collinear when
$k = -2$ ✓ ½

Answer $k = -2$

Watch out: Sign slip in $-9(3-k)$: it is $-27+9k$, not $-27-9k$.

19 **Long answer** Board-pattern 5 marks

Using vectors, show that the points $A(1,1,1)$, $B(2,3,3)$ and $C(3,2,-1)$ are the vertices of an isosceles right-angled triangle. Hence find the area of the triangle and the measure of $\angle B$.

Formula / idea: $\overrightarrow{AB}\cdot\overrightarrow{AC}=0$ means $\angle A=90^\circ$; area $=\frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$.

1 Find the side vectors.
$\overrightarrow{AB}=\hat{i}+2\hat{j}+2\hat{k},\quad \overrightarrow{AC}=2\hat{i}+\hat{j}-2\hat{k},\quad \overrightarrow{BC}=\hat{i}-\hat{j}-4\hat{k}$ ✓ 1

2 Find the lengths of the sides.
$|\overrightarrow{AB}|=\sqrt{1+4+4}=3,\quad |\overrightarrow{AC}|=\sqrt{4+1+4}=3,\quad |\overrightarrow{BC}|=\sqrt{1+1+16}=3\sqrt{2}$ ✓ ½

3 $AB=AC$, so the triangle is isosceles. ✓ ½

4 Find $\overrightarrow{AB}\cdot\overrightarrow{AC}$.
$\overrightarrow{AB}\cdot\overrightarrow{AC}=2+2-4=0\ \Rightarrow\ \angle A=90^\circ$ ✓ ½

5 Hence $\triangle ABC$ is an isosceles right-angled triangle. ✓ ½

6 Find $\overrightarrow{AB}\times\overrightarrow{AC}$ for the area.
$\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&2&2\\2&1&-2\end{vmatrix}=-6\hat{i}+6\hat{j}-3\hat{k},\quad |-6\hat{i}+6\hat{j}-3\hat{k}|=9$ ✓ ½

7 Halve it to get the area.
$\text{Area}=\frac{1}{2}\times 9=\frac{9}{2}\ \text{sq units}$ ✓ ½

8 For $\angle B$ use the two sides starting at B.
$\overrightarrow{BA}\cdot\overrightarrow{BC}=(-1)(1)+(-2)(-1)+(-2)(-4)=9$ ✓ ½

9 Find $\cos B$ and the angle.
$\cos B=\frac{9}{3\cdot 3\sqrt{2}}=\frac{1}{\sqrt{2}}\ \Rightarrow\ \angle B=\frac{\pi}{4}$ ✓ ½

Answer **Right angle at A, $AB=AC=3$; area $=\frac{9}{2}$ sq units; $\angle B=\frac{\pi}{4}$**

Watch out: Using $\overrightarrow{AB}$ and $\overrightarrow{BC}$ for $\angle B$; both vectors must start at B, so use $\overrightarrow{BA}$.

20 **Case study** Board-pattern 4 marks

Case Study: The triangular sail of a toy boat is a flat piece of cloth with corners O, P and Q. Taking O as the origin, $\overrightarrow{OP} = 2\hat{i} + 2\hat{j} + \hat{k}$ and $\overrightarrow{OQ} = 3\hat{i} + 4\hat{k}$ (units in metres).

(i) Find the length of the edge OP and the unit vector along $\overrightarrow{OP}$. 1 mark

(ii) Find the projection of $\overrightarrow{OQ}$ on $\overrightarrow{OP}$. 1 mark

(iii) Find the area of the sail. 2 marks

Formula / idea: Unit vector $= \dfrac{\vec{a}}{|\vec{a}|}$; projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$; triangle area $= \dfrac{1}{2}|\vec{a}\times\vec{b}|$.

(i) 1 mark

1 Length of OP is the magnitude of $\overrightarrow{OP}$.
$|\overrightarrow{OP}| = \sqrt{4+4+1} = 3$ m ✓ ½

2 Unit vector = vector ÷ its magnitude.
$\dfrac{\overrightarrow{OP}}{|\overrightarrow{OP}|} = \dfrac{1}{3}(2\hat{i} + 2\hat{j} + \hat{k})$ ✓ ½

Answer (i) $OP = 3$ **m; unit vector** $= \dfrac{1}{3}(2\hat{i} + 2\hat{j} + \hat{k})$

(ii) 1 mark

1 Find $\overrightarrow{OQ}\cdot\overrightarrow{OP}$.
$\overrightarrow{OQ}\cdot\overrightarrow{OP} = 3(2) + 0(2) + 4(1) = 10$ ✓ ½

2 Divide by $|\overrightarrow{OP}|$.
$\text{Projection} = \dfrac{10}{3}$ m ✓ ½

Answer (ii) $\dfrac{10}{3}$ **m**

(iii) 2 marks

1 Write the formula for the area.
$\text{Area} = \dfrac{1}{2}|\overrightarrow{OP}\times\overrightarrow{OQ}|$ ✓ ½

2 Find the cross product.
$\overrightarrow{OP}\times\overrightarrow{OQ} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ 3 & 0 & 4 \end{vmatrix} = 8\hat{i} - 5\hat{j} - 6\hat{k}$ ✓ ½

3 Find its magnitude.
$\sqrt{64 + 25 + 36} = \sqrt{125} = 5\sqrt{5}$ ✓ ½

4 Halve it to get the area of the sail.

$$\text{Area} = \frac{5\sqrt{5}}{2} \text{ sq m}$$

✓ ½

Answer (iii) $\frac{5\sqrt{5}}{2}$ **sq m**

Watch out: Dividing by $|\overrightarrow{OQ}| = 5$ in (ii); that gives the projection of $\overrightarrow{OP}$ on $\overrightarrow{OQ}$.